

Name: _____ Date: _____ Class: _____

MyDesign Portfolio

Element D Initial Template

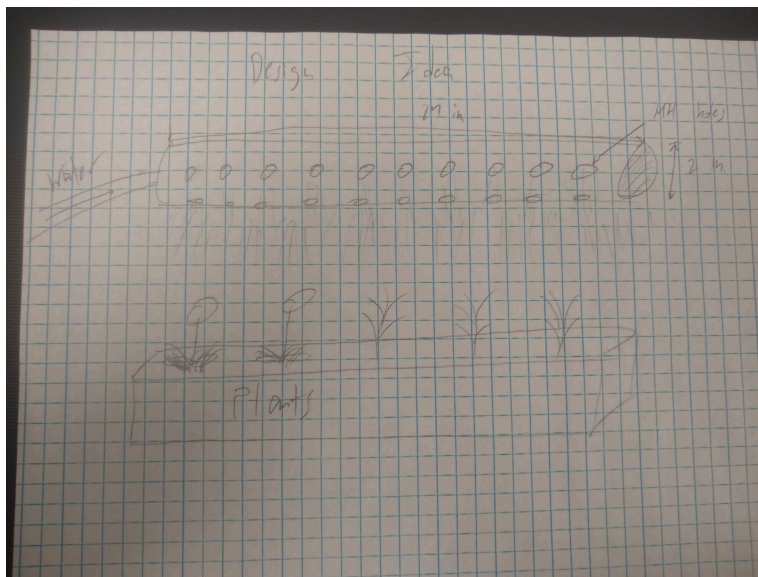
Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

Design Idea #1

Date: 3/11/25

Description:



Scale is 3 boxes to 2 inches.

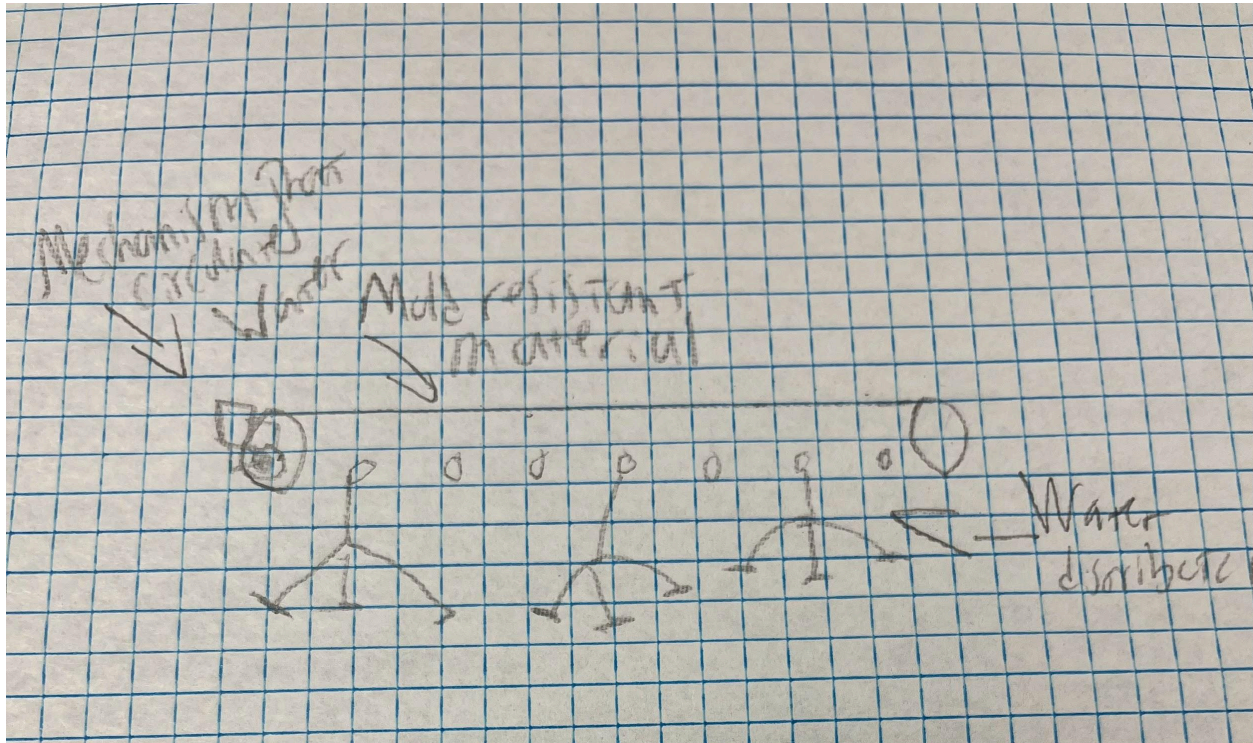
Pipe is made of PVC with M4 holes drilled in it

Tube is surgical tubing

Design Idea #2

Name: _____ Date: _____ Class: _____

Date: 3/12/2025



Scale is 4 boxes equals 1 foot

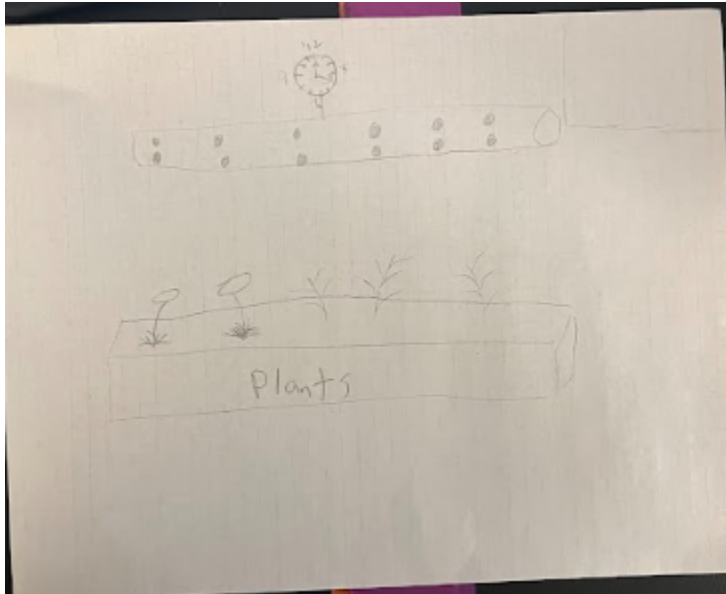
Description: A system similar to the current one however it is made out of mold resistant materials and has a mechanism which circulates the water preventing stagnant moisture which leads to mold growth.

Design Idea #3

Date: 3/13/2025

Description: automated scheduled watering system using a clock to schedule the watering times.

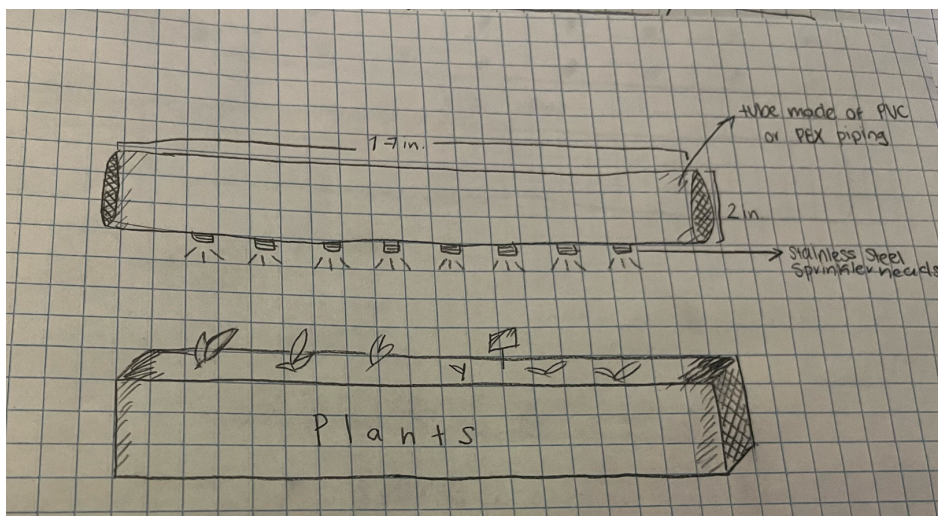
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Drawing that includes distinguishing information such as scale, dimensions, and materials:

Scale is 4 boxes equals 1 foot. Pipe is made out of PVC with M4 holes drilled into it. Tube is surgical tubing and the clock is a CASIO wrist watch.

Design Idea #4



Name: _____ Date: _____ Class: _____

Date: 3/11/25

Description: Scale is 3 boxes to 2 in. , includes a PVC pipe of 17 in. with stainless steel sprinkler heads on the bottom of the pipe going onto the plants.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Design Idea Comparison and Selection (*D1, D2*)

	Idea 1	Idea 2	Idea 3	Idea 4
lightweight	0	0	0	0
Low cost	1	1	0	0
anti-mold	1	1	1	1
Low maintenance	0	0	1	0
aesthetic	0	0	0	1
score	2	2	2	2

Our team chooses design 1 to solve the community problem because it is able to avoid the mold, and is low-cost unlike idea 3 and 4.

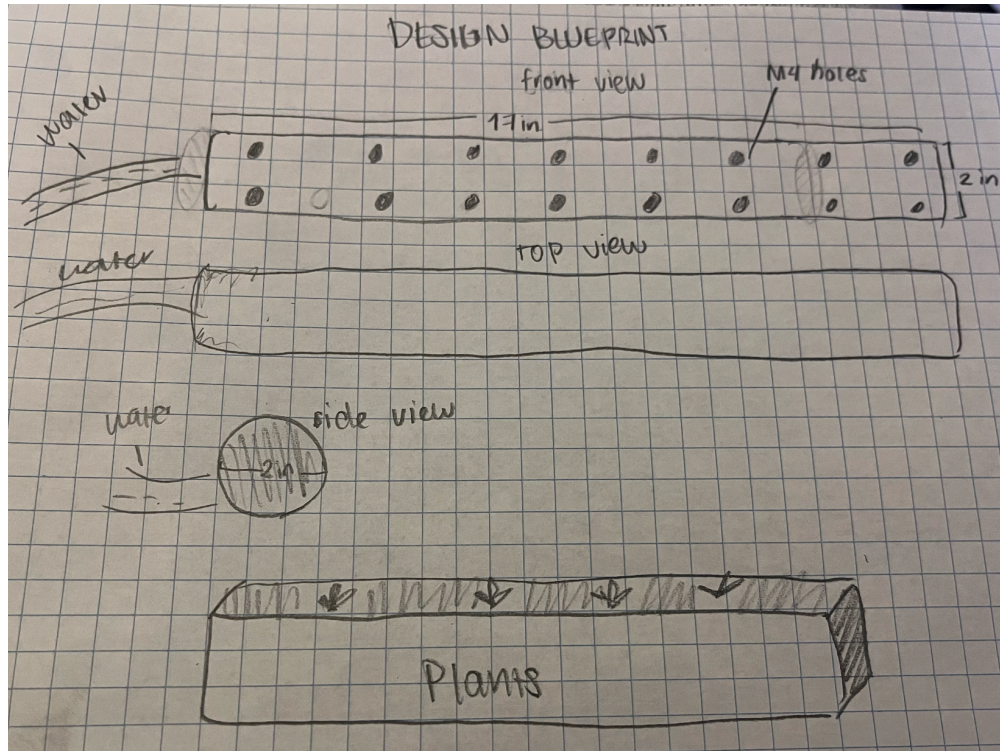
Design Blueprint (*D4*)

Design Blueprint #1 (Must have front view, top view, side view, and dimensions)

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Date: **3/11/25**

Sketch or CAD Drawing:



Scale: 3 boxes to 2 in.

Manufacturing Methods:

Step 1: **Measure the length of the PVC pipe, determine the diameter of the surgical tubing and ensure it fits comfortably inside the PVC pipe**

Step 2: **Cut the PVC to your liking using a PVC pipe cutter or saw and smooth the edges of the cut PVC pipe.**

Step 3: **Measure and Mark the Hole Positions and drill the M4 holes.**

Supply List and Costs:

Item	Cost each	# needed	Total
PVC Pipe	\$2 per foot	2 ft	Around \$4



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Surgical tubing	½ diameter, 2 ft - \$4	2 ft	Around \$4
M4 Drill Bits	\$3	Pack of 10	\$3
Total Cost			Around \$11

Plan of Action (D3)

Team #1 Gantt Chart:  **Copy of Gantt chart Team 1**

Iterations for the Solution:

5/6

We want to add sprinklers to spread the water out more, right now it is just going in a stream. It would be better to evenly distribute the water so plants are watered better. The sprinklers will be 3D printed, and we'll test multiple different sprinklers.

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The	was <i>comprehensive,</i> <i>iterative,</i> <i>and</i>	was <i>thorough,</i> <i>iterative,</i> <i>and</i> generally	was <i>adequate,</i> <i>generally</i> iterative and defensible	was <i>partial or</i> <i>overly general</i> <i>and only</i> <i>somewhat</i> iterative	was <i>incomplete</i> <i>or minimally</i> <i>iterative</i> <i>and/or</i>	was not evident <i>that there was</i> <i>an attempt</i>



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Process for Generating Possible Solutions _____.	<i>consistently</i> defensible (viable design highly likely)	defensible (viable design likely)	<i>(viable design is possible)</i>	and/or defensible (raising issues with the design viability)	<i>defensible (raising issues with the design viability)</i>	
2.Design Solution Choice. The Design Solution chosen _____.	was <i>well-justified</i> and demonstrated attention to <i>all</i> design requirements	was <i>justified</i> and demonstrated attention to <i>most if not all</i> design requirements	was <i>explained with reference to at least some</i> design requirements	was <i>not sufficiently explained with reference to</i> design requirements	<i>lacked support related to</i> design requirements	<i>was not attempted or evidence not provided</i>
3. Plan of Action. The plan of action _____.	<i>has considerable merit and would easily</i> support repetition and testing for effectiveness by others	would support repetition and testing for effectiveness by others	<i>might not clearly or fully support repetition and testing for effectiveness by others</i>	<i>has insufficient detail to allow for testing for replication of results</i>	<i>is lacking to the point that the design solution cannot be tested</i>	<i>cannot be executed or does not exist</i>